

Sampling Rate Conversion

Verifying an Interpolation Factor Claim

Claim: Interpolation factor $L = 5$ turns an 8 kHz input into an 18 kHz output

Question (10 marks): Verify the calculation and debug the error

The Interpolation Rule

$$F_s(\text{out}) = F_s(\text{in}) \times L$$

L = interpolation factor • L – 1 zero samples inserted between every original sample

Why the rate rises

Inserting zeros between samples raises the sample count without changing the time duration → the effective sampling rate scales up by L.

Time-domain check

Duration stays fixed. N samples at $F_s(\text{in})$ become $N \times L$ samples in the same time → new rate is $F_s(\text{in}) \times L$.

Follow-up step

A low-pass filter (interpolation filter) then removes the spectral images created by the zero-insertion.

The Stated Problem

Given

Input sampling frequency $F_{s(in)}$

8 kHz

Interpolation factor L

5

Claimed output frequency

18 kHz

Sanity Check

18 kHz is not an integer multiple of 8 kHz.

$$18 / 8 = 2.25$$

But an interpolation factor L must be a positive integer — the operation only inserts whole zero-samples between existing samples. A non-integer relationship between input and output rate is impossible for pure interpolation.

Correct Calculation

$$F_s(\text{out}) = F_s(\text{in}) \times L = 8 \text{ kHz} \times 5 = \mathbf{40 \text{ kHz}}$$

Step-by-step

- Start: 8000 samples/sec at the input
- Insert $L-1 = 4$ zero samples after every original sample
- New spacing between samples shrinks to $1/5$ of the original
- New rate = $5 \times 8000 = 40,000$ samples/sec = 40 kHz

Result

$F_s(\text{out}) = 40 \text{ kHz}$ is the only value consistent with an interpolation factor of 5 on an 8 kHz input.

$40 / 8 = 5 \checkmark$ (integer, matches L)

$18 / 8 = 2.25 \times$ (not an integer)

Debugging the Error

Likely mistake 1

$F_s(\text{in}) + F_s(\text{in}) \times L - 5 \times 8$ misread — possibly added instead of multiplied, or a wrong factor (e.g. 2.25) was used in place of 5.

Likely mistake 2

Confusing interpolation (rate $\times L$, up-sampling) with a filter cutoff or bandwidth figure, which is unrelated to the output sample rate.

Correct concept

Interpolation always scales the rate by an integer L . Decimation (down-sampling) instead divides: $F_s(\text{out}) = F_s(\text{in})/M$.

Conclusion

The stated result of 18 kHz is INCORRECT. For an interpolation factor of 5 applied to an 8 kHz input, the verified output sampling frequency is 40 kHz ($F_s(\text{out}) = F_s(\text{in}) \times L = 8 \times 5$).